Gautham Ramkumar

Ashburn, VA | +1 (857) 565-5374 | ramkumar.g@northeastern.edu
linkedin.com/in/rgautham20 | github.com/gautham-ramkumar | gautham-ramkumar.github.io
Authorized to work in the US for 3 years with no sponsorship required

KEY ACHIEVEMENTS

- Placed 2nd at Revolute Hackathon, shipping a 3-model voice-to-action manipulation pipeline in one weekend

EDUCATION

Northeastern University, Boston, MA May 2026
Master of Science in Robotics (Concentration: ECE) *GPA: 3.636/4.0*

Vellore Institute of Technology, India May 2024
Bachelor of Technology in Electronics and Communications Engineering *GPA: 3.48/4.0*

TECHNICAL SKILLS

Languages	C++17, Python
Deep Learning	PyTorch, CNNs / U-Net, Vision-Language Models, TensorRT, Edge Deployment
Vision & 3D	3D Gaussian Splatting, Structure from Motion, Bundle Adjustment, COLMAP, OpenCV
Hardware	NVIDIA Jetson Orin Nano, LiDAR, IMU, GNSS, SocketCAN, 6-DoF Arm
Software Practices	ROS2, ROS1 (Kalibr), Docker, Linux, Git, PyTest, GitHub Actions CI/CD, MuJoCo

PROFESSIONAL EXPERIENCE

Chicago Education Advocacy Corporation Aug 2026 – Present
Robotics AI Engineer, Chicago, IL (Remote)

- Grew automated test coverage from 44 to 145 tests across a two-job CI pipeline
- Reproduced a 70% SmolVLA task-success baseline in MuJoCo on an 8 GB GPU with pinned dependencies
- Built a dataset verifier that catches 3 silent defects (zeroed chunks, wrong state convention, no unnormalization)
- Built Backstop, a VLA failure monitor reaching 85.4% detection at 2.9% false-stop on held-out episodes

SmartInternz May 2023 – Jul 2023
Internet of Things Intern, Chennai, India

- Maintained 1 cm distance accuracy in a real-time ultrasonic sensing pipeline for automated access control

PROJECTS

DarkSight — Low-Light Perception 2025 – Present
PyTorch, TensorRT, RAFT, YOLOv8, RealSense | Code

- Designed a DS3DConv encoder-decoder with a ConvGRU bottleneck for frame memory and RAFT optical-flow warping for temporal consistency
- Reached 17.25 dB PSNR at 0.99 ORB stability, running 21 FPS at 360p with 57 ms latency on an Intel RealSense D435

YesChef: Voice-Controlled Robotic Sandwich Assembly Aug 2026
Python, LeRobot, Jetson Orin Nano, SocketCAN | Demo

- Added bounded retry logic around CAN control-ack timeouts, stopping crashes on the 6-DoF arm's control loop
- Fixed a leader-arm unwrap fault, rescaling the gripper so 270° of follower range spans 90° of leader travel

3D Gaussian Splatting from Scratch Jan 2026 – Mar 2026
Python, PyTorch, CUDA, COLMAP, Docker | Code

- Implemented the full Gaussian optimization pipeline from scratch (densification, cloning, splitting, pruning) on the official EWA CUDA rasterizer
- Established 7,500 iterations as the optimal stopping point for small scenes at 31.42 dB PSNR / 0.9531 SSIM, over 16 dB better than the 30k checkpoint

Low-Light Image Enhancement with U-Net 2025
Python, PyTorch, OpenCV | Code

- Designed a U-Net pipeline for joint denoising and exposure correction on the LOLv1 benchmark
- Reached 19.37 dB PSNR, a 16.9% improvement over the 16.57 dB baseline